

CSCN8020 Assignment 1 - Reinforcement Learning Programming

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Submission Type	One-page PDF with public GitHub repository link

100-word Summary of Work Completed

This assignment implements four reinforcement learning tasks in one organized Jupyter Notebook. I first model a pick-and-place robot as a Markov Decision Process by defining states, actions, rewards, and the reason each choice supports fast, smooth robot motion. I then solve a 2x2 Gridworld with two manual value-iteration updates and code verification. For the 5x5 Gridworld, I implement standard and in-place value iteration, compare convergence behavior, and report optimal values and policies. Finally, I implement off-policy Monte Carlo control with importance sampling and compare its estimates with dynamic programming results using runtime, episodes, and complexity observations.

Public GitHub Repository

https://github.com/chooksemmanuel/CSCN8020_Assignment1.git

Repository Contents

The repository contains one runnable Jupyter Notebook, object-oriented source files under **src/**, a sample execution log, README.md, requirements.txt, .gitignore, and supporting folders. The notebook includes Markdown explanations, manual calculations, Python implementation, output tables, policies, algorithm comparisons, and talking points for each problem.

Submission Reminder

Only this one-page PDF is submitted to Brightspace. The evaluator should clone the public GitHub repository using the URL above, which ends in **.git**.